Network Performance Analysis — Tutorial 2
$M/M/1$ Queue, Little's Formula

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1. The simple $M/M/1$ queueing system uses exponential probability densities to model the inter-arrival times and service times. The exponential density for inter-arrival times with parameter λ is given by the expression

$$p(t) = \lambda e^{-\lambda t}$$

For $t \geq 0$.

   (a) Sketch this density function.

   (b) Show that both the mean and standard deviation of this density are $\frac{1}{\lambda}$.

   (c) Briefly explain what these two properties of a probability density represent.

   (d) What is meant by the *memoryless* property of the exponential distribution?

2. In the lecture notes we saw that the average number of customers in the system for an $M/M/1$ queue was given by

$$\langle k \rangle = \frac{\rho}{1-\rho}$$

show that the variance of k is given by

$$\left\langle k^2 \right\rangle - \langle k \rangle^2 = \frac{\rho}{(1-\rho)^2}.$$

3. For an $M/M/1$ queue sketch curves of the following quantities vs ρ, the utilization factor.

   (a) The average number of customers in the system.

   (b) The average time delay of the system.

   (c) Using these plots, comment on the consequences of allowing ρ to approach 1.

4. Students arrive randomly, in a Poisson distributed manner, at a cash-dispensing machine with an average inter-arrival time of 5 minutes. The length of time that a student spends on the machine is exponentially distributed with an average of 2 minutes.

(a) What is the probability that a student arriving at the machine will have to wait?

(b) What is the average waiting (queueing) time for a student?

(c) The bank plans to install a second machine when the average waiting time is 5 minutes or longer. At what average arrival rate will this occur? For this arrival rate, what is the *total* average delay time that students spend at the machine?

(d) What is the average queue length for the arrival rate of 4c?

5. For a system with a single server (such as the $M/M/1$ and $M/M/1/m$ queue) the utilisation factor, ρ, is the probability that the queue state is not zero. For a lossless system we saw, using Little's formula (which is valid for $M/G/1$ queues), that the utilisation factor is given by $\rho = \frac{\lambda}{\mu}$. Use the stationary distribution for the $M/M/1/m$ queue to compute the utilisation factor for the $M/M/1/m$ queue. Explain why it is not $\frac{\lambda}{\mu}$.